CS4240/6240 Fall 2026 Course Project Description

Your project is worth 60 points. *This is a small security related project, not Master's thesis size project. It has to be practical and interesting and/or fun. You will upload it to the github and share link with the class.*

The goal of the project is to further develop understanding of the course material, which is centered around system administration and security. Therefore, your project should explore topics related to security. Projects are done in groups of up to 4 students.

Students will need to conduct research into similar projects and motivate why their project is important. This project should become a well-documented open source project that others in the Linux community can use. It is recommended that the project is housed on github, but it is not required. Project can be based on other open source projects. You can use any language of your choice, if your project calls for writing code. The general topic areas are listed below:

- Develop an open source tool that detects intrusion to the system, suspends the attack, and secures the system.
- Develop a visual or command-line tool to penetrate a system (geared towards the Ubuntu Linux distribution).
- Develop a tool to collect and process data from IoT.

You are welcome to propose topics of your choice, BUT all topics are subject to approval! Projects have to be unique and two groups will not be presenting the same topic, so send in the proposal the moment you decide what you want to do.

Project Proposal Due: 10/16 before 11:59 pm (0 points):

Write a 1-page (or less) *informal* project proposal in .pdf format. Clearly identify the objectives of your project. Describe in detail what you are doing and how you are doing it. Explain briefly why your project is important/cool/useful/etc. List members of your group (no more than 4), and describe the system/s that you are using for your project. Your project should either use a VM, or any Linux installation of your choice, *not* Windows and *not* Mac. Submit to canvas to get your project approval and feedback.

Your project will be publicly available to students in the class, so do not use any topics that are classified, or cannot be publicly shared. Also, the demo will be done LIVE in class, so video recordings, home servers, or pre-collected data will not be acceptable.

Presentation in class on 11/17, 11/19 and 11/24:

You will present your project in class in person, live (no recordings). Your presentation should be an average of 10-12 minutes. Provide motivation why you have done what you have done, and why it is important/cool/fun/etc. You would need about 10 slides (not more than that) not including the title page and references just to show the highlights of your project. No more than 15 slides.